

using Idk dataset, the number of IK-IK and IK-IDK has significantly decreased, indicating that the model's ability to be aware of its own knowledge has been enhanced. However, the model may also refuse to answer some questions it knows, leading to a decrease in the number of IK-IK questions. Compared to SFT model, preference-aware optimization (like DPO) can mitigate the phenomenon where the model incorrectly refuses to answer questions it knows. Besides, we conduct extensive ablation experiments to explore the effect of Ik threshold, data sources, model size and other settings.

Our findings can be summarized as follows:

1. After aligning using Idk datasets, AI assistants are capable of largely knowing what they know and what they do not know and refusing their unknown questions. Llama-2-7b-chat can definitively determine whether it knows the answer to up to **78.96%** of the questions in the test set. And it exhibits good performance on out-of-distribution test sets.
2. Supervised fine-tuning cause the model to become overly conservative, incorrectly rejecting known questions. Preference-aware optimization can mitigate this problem, promoting the overall proportion of IK-IK and IK-IDK questions.
3. The Ik threshold used to define knowns and unknowns questions influences the behavior of the assistant. The more questions labeled as "I don't know," the more likely the assistant is to refuse to answer questions. In general, the higher the Ik threshold, the greater the total number of Ik-IK and Ik-IDK questions, resulting in a more truthful assistant.
4. Larger model is more adept at distinguishing which questions it knows and which it doesn't know. The use of Idk-SFT on Llama-2-70b-chat, as compared to Llama-2-7b-chat, results in a 5.8% improvement in the total number of IK-IK and IK-IDK questions.

## 2 Background

### 2.1 Aligning LLMs with Human Values

To build AI assistants based on large language models, we typically need to align these large language models with human values, making them helpful, truthful and harmless (Askell et al., 2021; Bai et al., 2022; Ouyang et al., 2022). Here we introduce several mainstream alignment methods related to our work. The most common alignment method for pre-trained models is instruction tuning, also known as Supervised Fine-Tuning (SFT). Wei et al. (2022a); Sanh et al. (2022) fine-tune pre-trained models on a collection of NLP datasets combined with natural language instructions to enhance zero-shot performance on unseen tasks. Chung et al. (2022); Longpre et al. (2023) scale the number of tasks, the model size and fine-tune on mixed data. Sun et al. (2023) utilize Self-Instruct (Wang et al., 2023c) to synthesize three types of SFT data - helpful, honest, and harmless - and construct a conversational assistant. The step following SFT is preference optimization. Bai et al. (2022); Ouyang et al. (2022) use Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Stiennon et al., 2020). They first train a reward model on the human preference data and then optimize the policy model using Proximal Policy Optimization (PPO) (Schulman et al., 2017) with the trained reward model. Zhang et al. (2023a) propose a reward-free method named Hindsight Instruction Relabeling (HIR) to utilize preference data by converting feedback to instructions and training the model using supervised fine-tuning. Rafailov et al. (2023) propose Direct Preference Optimization (DPO) which can directly fine-tune language models to align with human preferences without the need of reward modeling.

### 2.2 Discovering LLMs' Knowledge

Large language models store extensive world knowledge during the pre-training. There has been increasing interest in researching knowledge in large language models. Kadavath et al. (2022); Lin et al. (2022b) fine-tune language models using a classification head or verbalized

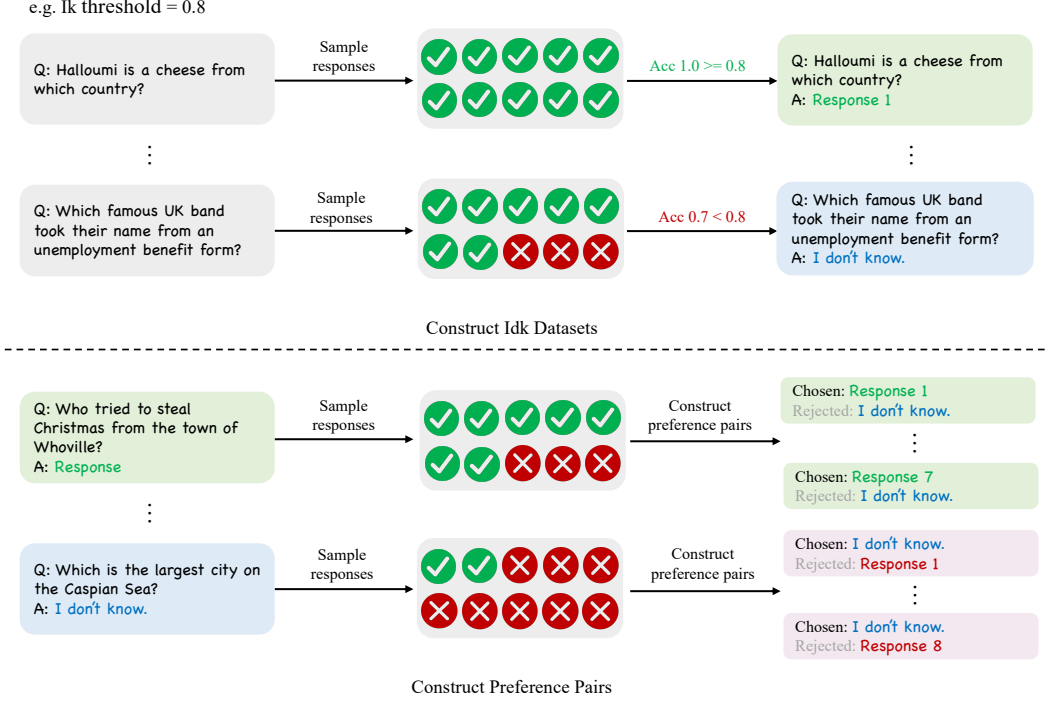


Figure 3: **Top:** Construction process of the Idk dataset. **Bottom:** Construction process of preference pairs. The green **response** indicates a correct answer, the red **response** indicates an incorrect answer, and “I don’t know” represents the template for refusal to answer.

confidence. However, they don’t teach models to aware their knowledge boundary and refuse to answer the questions they don’t know. Yin et al. (2023b) propose the SelfAware dataset to evaluate the ability of LLMs to recognize what they don’t know, finding that there is still an apparent gap compared to human. Burns et al. (2023) propose an unsupervised method to find latent knowledge inside the activations of a language model by answering yes-no questions given only unlabeled model activations. Ren et al. (2023) investigate whether LLMs can perceive their knowledge boundaries or not under retrieval-augmented setting and normal setting. Zhao et al. (2023); Manakul et al. (2023); Anonymous (2023) check the generated answers in an unsupervised way to judge whether the model knows the question. The basic idea is that if the model knows the answer to a question, then the diversity of multiple sampling generations should be relatively low.

### 2.3 Mitigating LLMs’ Factual Errors

There are some studies focused on eliminating factual errors in AI assistants. Asai et al. (2023) propose a framework named SELF-RAG to enhance an LM’s factuality by retrieval augmentation and self-reflection. Li et al. (2023) first find truthful directions through probing and then do inference-time intervention in these truthful directions. Zou et al. (2023) use representation engineering to enhance factuality of the model’s output. Chuang et al. (2023) propose a simple decoding strategy for reducing hallucinations by contrasting the differences in tokens’ logits obtained from different layers. Tian et al. (2023) directly fine-tune language models to learn factuality from preference dataset using direct preference optimization. However, there is currently no method that can guarantee the complete elimination of factual errors. In practical applications, it is a necessary feature for AI assistants to refuse to answer questions they do not know.

### 3 Methodology

#### 3.1 Construction of the Idk Dataset

Given a knowledge-intensive question answering dataset, it is hard to precisely determine which questions the model truly knows the answers to, which due to the model having varying degrees of confidence for different knowledge. Therefore, following the approach of previous work (Kadavath et al., 2022; Lin et al., 2022b), we sample multiple responses from the model for each question, calculating the accuracy rate across these responses as a measure of the model’s confidence regarding that question. Finally, we select a specific level of confidence as the criterion for determining whether the model knows or does not know the answer to a question, that is the Ik threshold. To construct the QA pairs in the Idk dataset, for questions that the model does not know, we use a template for refusal to reply as the answer. For questions that the model knows, we select a correct response generated by the model itself as the answer. The procedure is demonstrated in Figure 3 (top). Our refusal to answer template is:

This question is beyond the scope of my knowledge, and I am not sure what the answer is.

We use “I don’t know” and the Idk template to refer to this template in the following paper.

**Determine whether the output of a model is correct.** In order to construct the Idk dataset, we need an automatic method to evaluate whether the model’s outputs are correct. According to the experimental results presented in Wang et al. (2023a), employing lexical matching, which checks whether the golden answers appear in the responses generated by the model, to evaluate on a subset of the TriviaQA’s validation set (Joshi et al., 2017) yields a consistency rate of approximately 90% with human evaluations. We consider lexical matching to be a relatively accurate automatic evaluation method for the TriviaQA dataset. Besides, TriviaQA is a mainstream knowledge-intensive open-domain question answering dataset. Therefore, we construct our Idk dataset based on the TriviaQA dataset.

**Meaning of different Ik thresholds.** The model is required to refuse to answer questions where the confidence level is below the Ik threshold. It is important to note that different Ik thresholds will result in different Idk datasets. A high Ik threshold indicates that the model will only answer a question if it possesses a high level of confidence. Conversely, if the Ik threshold is low, the model will answer questions with a lower level of confidence required. In other words, a high Ik threshold represents a conservative response strategy, whereas a low Ik threshold represents a more aggressive response strategy. In this work, we sampled ten responses for each question and derived ten discrete Ik thresholds based on varying accuracy rates. For the sake of simplicity, we set the Ik threshold to 1.0, meaning that the model is considered to know the answer to the question only if all ten of its responses are correct. Unless specifically stated otherwise, the Idk dataset mentioned hereafter is constructed based on an Ik threshold of 1.0. We discuss the impact of different Ik thresholds in Section 4.

In the following sections, we introduce our methods to teach AI assistants to say “I don’t know” when encounter unknown questions. Since the AI assistant we discuss is based on large language models, we will interchangeably use the terms “model” and “assistant” in the following sections.

#### 3.2 Idk Prompting

For models capable of following human instructions, such as Llama-2-7b-chat, We can directly instruct an assistant to say “I don’t know” to unknown questions by adding a prompt in front of the input question. We call this method Idk-Prompting. This requires the model to have a high capability for following instructions, but the advantage is that it eliminates the need for additional training. We call such a prompt an Idk prompt. Our Idk prompt is as follows:

Answer the following question, and if you don’t know the answer, only reply with "I don’t know": <Question>